Student Guide — Working with JupyterLite

Welcome! In this course you will write and run Python directly **in your web browser**, using a tool called **JupyterLite**. There is **nothing to install**, **no account to create**, and **nothing to download**: when you open the course link, all the notebooks and data files are already there. This short guide tells you everything you need. Read it once before the first lesson.

Course links

Course home: https://andreatagliabue.github.io/python-unige/

This single link opens everything. You can also jump straight to a specific lesson:

Lesson	Link
1 — Introduction	https://andreatagliabue.github.io/python-unige/notebooks/index.html?path=01_introduction.ipynb
2 — Python basics	https://andreatagliabue.github.io/python-unige/notebooks/index.html?path=02_python_basics.ipynb
3 — Data I/O and tables	https://andreatagliabue.github.io/python-unige/notebooks/index.html?path=03_data_io_and_tables.ipynb
4 — Plotting and analysis	https://andreatagliabue.github.io/python-unige/notebooks/index.html?path=04_plotting_and_analysis.ipynb
5 — Fitting, statistics, automation	https://andreatagliabue.github.io/python-unige/notebooks/index.html?path=05_fitting_stats_automation.ipynb

> **Tip:** bookmark the course home so you can always find it. Open it in a **normal** browser window (not Private/Incognito — see Section 5).

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1. What is JupyterLite (and what you need)

- **JupyterLite** is a version of the Jupyter Notebook environment that runs entirely inside your browser. Python itself runs on your computer, *inside the browser tab* — not on a remote server.
- **You do NOT need to install Python, Anaconda, or anything else.**
- **You do NOT need to log in or create an account.**
- **You do NOT need to upload anything:** the course materials are already loaded for you.

What you DO need:

- A laptop (Windows, macOS, or Linux — it does not matter).
- A recent version of **Google Chrome, Microsoft Edge, or Firefox**. Avoid Internet Explorer and very old browsers.

- An **internet connection**, especially the first time you open it (the browser downloads Python and the scientific libraries the first time — this can take 30–60 seconds).
- Do **not** use the browser in *Private / Incognito* mode for the course: your work would be deleted every time you close the window (see Section 5).

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2. Opening the course

1. Open your browser.
2. Go to the **course home** (or pick a lesson from the **Course links** table at the top of this guide):

   > **https://andreatagliabue.github.io/python-unige/**

3. Wait a few seconds. You will see the **JupyterLab interface**:
   - on the **left**, a *file browser* showing the course notebooks and a `data` folder (these are already there for you);
   - in the **center**, the working area where notebooks open as tabs.

> **First time is slower.** The very first time you run Python code, the browser downloads the Python engine and libraries. This happens only once per browser; afterwards it is fast.

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3. Opening and running a notebook

1. In the **left file browser**, **double-click** the notebook for the lesson (a file ending in `.ipynb`). It opens as a tab in the center.
2. A notebook is a sequence of **cells**. There are two kinds:
   - **text cells** (explanations, in formatted text);
   - **code cells** (grey boxes containing Python).
3. To **run a cell**: click on it, then press **`Shift`** **+** **`Enter`**.
   - The code runs, the result appears just below the cell, and the selection moves to the next cell.
4. The small number in brackets to the left of a code cell, e.g. `[1]`, shows the order in which cells were run. While a cell is running you see `[*]`.

Run everything in order

The **order matters**: later cells often depend on earlier ones. The safest way to start a lesson is:

- Open the menu **`Run`** **→** **`Run All Cells`** to execute the whole notebook from top to bottom.

If something behaves strangely, reset and start clean:

- **`Kernel`** **→** **`Restart Kernel and Run All Cells…`** — this clears everything and runs the notebook again from the beginning.

> **Golden rule for beginners:** if you get confused, do **Restart Kernel and Run All Cells**. Most "it doesn't work" problems disappear after this.

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4. Editing and experimenting

- Click inside a code cell to edit the Python text, then press `Shift` **+** `Enter` to run it again.
- Add a new cell with the `+` button in the toolbar.
- It is completely normal to get **error messages** (red text). An error does not mean you are bad at this — it just means Python found something it could not understand. Read the **last line** of the error: it usually tells you what went wrong.

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5. Saving your work (please read — this is important!)

Your work lives **inside your browser's storage on this specific computer**. This has two consequences:

- ✓ If you simply close and reopen the tab on the **same browser and same computer**, your work is still there.
- ✗ Your work is **NOT** synced to the cloud and is **NOT** available on another computer.
- ✗ If you clear your browser history / cached data, or use *Private / Incognito* mode, your work **will be lost**.

Therefore, at the end of every lesson, download a copy of your notebook to your computer:

1. Open the menu `File` **→** `Download` (or right-click the file in the left browser → **Download**).
2. Save the `.ipynb` file somewhere safe on your computer.

If you ever want to continue your downloaded notebook later, you can drag it back into the file browser to upload it.

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6. Troubleshooting

Problem	What to do
`NameError: name '...' is not defined`	You probably ran cells out of order, or skipped one. Do **Kernel → Restart Kernel and Run All Cells**.
First cell takes very long / seems stuck	The first run downloads Python and the libraries. Wait up to a minute and make sure you have internet.
A cell shows `[*]` and never finishes	It may be waiting or stuck. Use **Kernel → Interrupt Kernel**, then run again.
Everything looks broken (errors everywhere)	This is almost always the *running state*, not your file. Do **Kernel → Restart Kernel and Run All Cells** — reloading the page also resets the running state. This fixes the large majority of issues.
I edited the notebook and want the ORIGINAL back	Your saved edits live in the browser, so **a reload keeps them** (it does not undo your changes). To restore the original course version, clear this site's data in your browser (browser settings → *clear site data / browsing data* for this site), then reopen the link. Files you never edited are always the originals.
`FileNotFoundError` **when reading a data file**	Make sure you are running the notebook from the course link (where the `data` folder is provided). If it persists, reload the page and tell the instructor.

7. Quick reference

- **Run a cell:** `Shift` + `Enter`
- **Run the whole notebook:** `Run` → `Run All Cells`
- **Clean restart:** `Kernel` → `Restart Kernel and Run All Cells…`
- **Save to your computer:** `File` → `Download`
- **Add a cell:** `+` button in the toolbar

You are ready. See you in the first lesson!